WORKED SOLUTION

STEP I 1991, Question 1

Pure • Trigonometric Identities

> **STEP I 1991, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/91-s1-q1

The key idea

The condition on the angles lets us remove one of them: since $\psi = \frac{1}{2}\pi - (\theta + \phi)$, we have $\sin\psi = \cos(\theta + \phi)$, and the identity becomes a statement about two free angles. After that, the products $\sin\theta\sin\phi$ and $\cos(\theta + \phi)$ combine to give $\cos(\theta - \phi)$, and the product $\cos(\theta + \phi)\cos(\theta - \phi)$ has a very tidy form. For the second part, the special choice of angle makes ψ equal to *half* of θ, which is where the information about $\frac{1}{10}\pi$ comes from.

Solution

The identity

Since $\theta + \phi + \psi = \frac{1}{2}\pi$,

$$\sin\psi = \sin(\tfrac{1}{2}\pi - (\theta + \phi)) = \cos(\theta + \phi).$$

So the left-hand side of the identity is

$$L = \sin^2\theta + \sin^2\phi + \cos(\theta + \phi)[\cos(\theta + \phi) + 2\sin\theta\sin\phi].$$

Inside the bracket, $\cos(\theta + \phi) = \cos\theta\cos\phi - \sin\theta\sin\phi$, so

$$\cos(\theta + \phi) + 2\sin\theta\sin\phi = \cos\theta\cos\phi + \sin\theta\sin\phi = \cos(\theta - \phi).$$

Now we need the product $\cos(\theta + \phi)\cos(\theta - \phi)$. Multiplying out the compound-angle formulae,

$$\begin{aligned}\cos(\theta + \phi)\cos(\theta - \phi) &= (\cos\theta\cos\phi - \sin\theta\sin\phi)(\cos\theta\cos\phi + \sin\theta\sin\phi)\\ &= \cos^2\theta\cos^2\phi - \sin^2\theta\sin^2\phi\\ &= \cos^2\theta\,(1 - \sin^2\phi) - (1 - \cos^2\theta)\sin^2\phi\\ &= \cos^2\theta - \sin^2\phi.\end{aligned}$$

Therefore

$$L = \sin^2\theta + \sin^2\phi + \cos^2\theta - \sin^2\phi = \sin^2\theta + \cos^2\theta = 1,$$

which is the required identity.

The cubic for $\sin\frac{1}{10}\pi$

Take $\theta = \phi = \frac{1}{5}\pi$. Then $\psi = \frac{1}{2}\pi - \frac{2}{5}\pi = \frac{1}{10}\pi$. Write

$$x = \sin\tfrac{1}{10}\pi, \qquad s = \sin\tfrac{1}{5}\pi.$$

The identity gives

$$2s^2 + x^2 + 2s^2x = 1, \qquad \text{that is,} \qquad 2s^2(1+x) = 1 - x^2. \tag{$*$}$$

Ask yourself Why not just use $\sin\psi = \cos 2\theta = 1 - 2s^2$ to eliminate s?

That relation holds for *every* θ with $\psi = \frac{1}{2}\pi - 2\theta$, so substituting $s^2 = \frac{1}{2}(1-x)$ into $(*)$ gives $(1-x)(1+x) = 1 - x^2$: true, but empty. We have to use something special about $\frac{1}{5}\pi$. The special fact is that $\frac{1}{5}\pi = 2\cdot\frac{1}{10}\pi$, so $\theta = 2\psi$, and the double-angle formula ties s to x:

$$s = \sin 2\psi = 2\sin\psi\cos\psi, \qquad \text{so} \qquad s^2 = 4x^2(1-x^2).$$

Substituting this into $(*)$,

$$8x^2(1-x^2)(1+x) = 1 - x^2.$$

Since $0 < \frac{1}{10}\pi < \frac{1}{2}\pi$, we have $0 < x < 1$, so $1 - x^2 \neq 0$ and we may divide by it:

$$8x^2(1+x) = 1, \qquad \text{that is,} \qquad 8x^3 + 8x^2 - 1 = 0.$$

So $\sin\frac{1}{10}\pi$ satisfies the given cubic, as required.

Checks

The identity can be tested with $\theta = \frac{1}{2}\pi$, $\phi = \psi = 0$: the left-hand side is $1+0+0+0 = 1$. With $\theta = \phi = \psi = \frac{1}{6}\pi$ it is $3\cdot\frac{1}{4} + 2\cdot\frac{1}{8} = 1$.

For the cubic, $\sin\frac{1}{10}\pi = \sin 18^\circ \approx 0.309017$, and $8x^3 + 8x^2 - 1$ evaluates to 0 to within rounding there. In fact the cubic factorises as $(2x+1)(4x^2+2x-1)$, with roots $-\frac{1}{2}$ and $\frac{1}{4}(-1 \pm \sqrt{5})$, and $\frac{1}{4}(\sqrt{5}-1) \approx 0.309017$ is the familiar exact value of $\sin 18^\circ$.

Common slips

- Eliminating s with $\sin\psi = \cos 2\theta$ only, and getting the identity $0 = 0$; see the box above.
- Dividing by $1 - x^2$ without saying why it is nonzero. Here $x = \sin\frac{1}{10}\pi$ lies strictly between 0 and 1.

Beyond STEP

The other two roots of the cubic are not accidents: $-\frac{1}{2} = \sin\left(-\frac{1}{6}\pi\right)$ and $\frac{1}{4}(-1-\sqrt{5}) = -\sin\frac{3}{10}\pi$. Equations like this one record the algebraic relations between values of sin and cos at rational multiples of π, which are best understood through roots of unity: $\cos\frac{1}{5}\pi$ is closely tied to the fifth roots of unity, the solutions of $z^5 = 1$. That is why constructing a regular pentagon with ruler and compasses comes down to $\sqrt{5}$. Needham's *Visual Complex Analysis* is a beautiful place to see these ideas geometrically.

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